CHAPTER 9

The exponential function and continuous models

Chapter path

Objectives. Use the algebraic properties; differentiate; study growth or decay; link the exponential to geometric sequences.
Prerequisites. Powers, geometric sequences, differentiation and sign.
Yearly reference point. 2–11 February: 8 hours

Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Simplify $2^3 \times 2^4$.
2. Express 3^{-2}.
3. Solve $2x + 1 = 5$.
4. Differentiate $x^2 + 2x$.
5. A 20 % decrease gives which coefficient?
6. For a geometric sequence $u_0 = 5, q = 2$, calculate u_n.

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Explain the difference between a continuous rate and a discrete multiplying coefficient; present an inequality proved through the minimum of a difference.

Investigate before formalising

Compare the increments of e^x and the values of the function around 0, with several positive steps. Build three Euler steps and explain the rule chosen.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

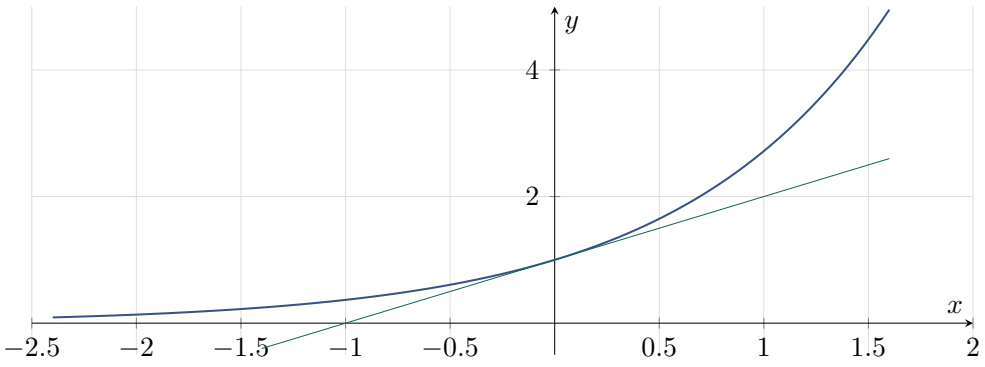
9.1 Defining a new function

Definition and results

The exponential function is the unique function f differentiable on $\mathbb{R}$ satisfying $f' = f$ and $f(0) = 1$. Its existence and uniqueness are accepted. It is written exp, then e^x with $e = \exp(1) \approx 2.71828$. The number e is a constant; x is the variable. The exponential is strictly positive.

Status and justification. Existence and uniqueness accepted in accordance with the programme. Positivity is deduced, after the algebraic properties accepted below, from $\exp(x) = \exp(\frac{x}{2})^2$ and from $\exp(x)\exp(-x) = 1$.

Worked example. $\exp(0) = 1$, $\exp(1) = e$, $\exp'(0) = 1$. The tangent at 0 is therefore $y = 1 + x$. The notation e^x extends the properties of powers to every real exponent.



e^x (blue) and its tangent $1 + x$ at 0 (green).

Watch out

$e^0 = 1$, not 0. The function never takes the value 0, even for negative x.

Immediate application. What is the equation of the tangent to e^x at 1?

9.2 Algebraic properties and comparisons

Definition and results

For all real x,y, $e^{x+y} = e^x e^y$, $e^{-x} = \dfrac{1}{e^x}$, $e^{x-y} = \dfrac{e^x}{e^y}$. For integer n, $(e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, so $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

Status and justification. The addition relation is accepted in the common pathway; it can be explored further via uniqueness. Positivity and $\exp' = \exp > 0$ give strict increase. This justifies the equivalences for equations and inequalities.

Worked example. $e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ is equivalent to $3x - 2 = x + 4$, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

Watch out

e^{x+y} is a product, in general never $e^x + e^y$. We do not solve $e^x = 2$ with a logarithm, which is not studied in Première.

Immediate application. Simplify $\dfrac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

9.3 Differentiating and studying

Definition and results

For real a, $t \mapsto e^{at}$ is differentiable and its derivative is ae^{at}. For $e^{at+b} = e^b e^{at}$, the derivative is ae^{at+b}. The product and quotient rules apply with the derivatives already known. The sign of the exponential often simplifies the study of the sign of the derivative.

Status and justification. The formula for differentiating e^{at} is accepted; the case with b follows by taking out the constant factor e^b. The general chain rule is not needed here.

Worked example. For $f(x) = (x + 1)e^{-x}$, $f'(x) = e^{-x} - (x + 1)e^{-x} = -xe^{-x}$. Since $e^{-x} > 0$, the derivative has the sign of $-x$: global maximum 1 at 0.

Watch out

Do not forget the factor a or the negative sign in e^{-x}. The exponential factor is positive, not necessarily increasing.

Immediate application. Differentiate $g(t) = 3e^{-2t}$ and study its direction of variation.

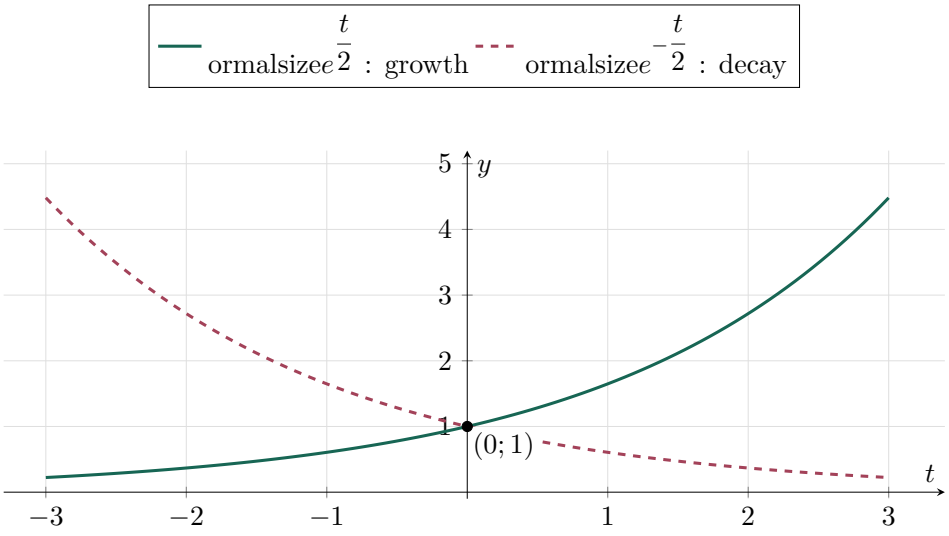
9.4 From discrete to continuous

Definition and results

A model $Q(t) = Q_0 e^{kt}$, with $Q_0 > 0$, represents growth if $k > 0$, decay if $k < 0$, constancy if $k = 0$. For a fixed step h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. The values at integer times form a geometric sequence with common ratio e^k. The unit of k is the inverse of the unit of time.

Status and justification. The ratio follows from $e^{k(t+h)} = e^{kt}e^{kh}$. The direction of variation is deduced from $Q\prime(t) = kQ_0e^{kt}$ and the positivity of the remaining factor.

Worked example. $Q(t) = 100e^{-0.2t}$, t in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0.2}$. A simulation can look for the first whole number of hours at which Q falls below 50; the exact equation is not solved with a logarithm in this chapter.



The curves for $k = \dfrac{1}{2}$ and $k = -\dfrac{1}{2}$ are symmetric about the vertical axis. The line styles also distinguish them without colour.

Watch out

The relative rate per discrete unit of time is $e^k - 1$, not exactly k; they are close only for small k.

Immediate application. For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

9.5 Building numerically and proving an inequality

Definition and results

To approximate the solution of $y' = y$, $y(0) = 1$ with a step h, Euler replaces the increment by hy: $y_{k+1} = (1+h)y_k$. On $[0;1]$ with $h = \dfrac{1}{n}$, the approximation of e is $\left(1+\dfrac{1}{n}\right)^n$. This construction is experimental. The inequality $e^x \geq 1 + x$ is obtained by studying a function.

Status and justification. For $g(x) = e^x - 1 - x$, $g'(x) = e^x - 1$, negative if x<0 and positive if x>0. The minimum is $g(0) = 0$, which proves the inequality; equality only at 0. The convergence of Euler's method is not proved here.

Worked example. With n=2, Euler gives $y_0 = 1, y_1 = 1.5, y_2 = 2.25$. With n=10, $(1.1)^{10} \approx 2.59374$. We compare with e; we do not confuse an approximate value with the definition of the function.

Watch out

A graph showing a curve above a line does not prove the inequality for all real numbers.

Immediate application. Show that $e^{2x} \geq 1 + 2x$ for every real number x.

Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 9.1 – Transforming

For every real number x, consider the expressions

$$A = e^x e^{2-x}, \qquad B = \frac{(e^{2x})^3}{e^x}, \qquad C = e^{-x}(e^x + 1).$$

1. Simplify A and B, justifying each grouping of exponents.
2. Expand then simplify C. Explain why a sum does not simplify in the same way as a product.

Exercise 9.2 – Equations and inequalities

Consider an equation and an inequality in the real unknown x:

$$e^{2x-1} = e^{x+3} \qquad \text{and} \qquad e^{-3x} \geq e^{x-2}.$$

1. Solve each in $\mathbb{R}$, citing the property of the exponential that allows the exponents to be compared.
2. While solving the inequality, explain at which step the direction of the inequality changes. Give the solution set as an interval.

Exercise 9.3 – Product and maximum

The function f is defined on $[0;4]$ by $f(x) = xe^{-x}$.

1. Calculate $f\prime(x)$, then study its sign on the interval.
2. Draw the variation table of f. Deduce its maximum and the x-coordinate where it is reached.
3. Check that the values at the endpoints do not give a larger maximum. Keep the exact values.

Exercise 9.4 – A modelled population

The following two parts are independent.

Part A: A population. A population is modelled by $P(t) = 200e^{0.1t}$, where $t \geq 0$ is the time elapsed in years. $P(t)$ is a theoretical count, which is not necessarily an integer.

1. Determine the initial count, then the multiplying coefficient between times t and $t+1$. Deduce the exact annual rate and its percentage rounded to the nearest hundredth.
2. Explain why this model cannot necessarily describe a real population indefinitely.
3. **Part B: Two curves.** In the same frame, sketch the curves of $t \mapsto e^{\frac{t}{2}}$ and $t \mapsto e^{-\frac{t}{2}}$. State their variation, their common point and their symmetry. Identify each curve clearly.

Exercise 9.5 – Searching for a threshold

The quantity of a substance, in milligrams, is modelled at time $t \geq 0$ by $Q(t) = 80e^{-\frac{t}{2}}$, where t is expressed in hours. Checks take place at integer times $n = 0, 1, 2, \ldots$. We look for the first check at which the quantity is strictly less than 10 mg.

1. Write an algorithm that searches for this first integer. Provide an iteration bound and state what the program returns if the bound is reached.
2. Run the algorithm, then check numerically the quantity at the index found and at the previous index.
3. Justify, using the decrease of Q, that this double check establishes that it is indeed the first index. Do not use a logarithm.

Exercise 9.6 – A symmetry

Consider the function f defined on $\mathbb{R}$ by $f(x) = e^{x} + e^{-x}$.

1. Study the parity of f and interpret the result for its curve.
2. Prove that $f(x) \geq 2$ for every real number x, choosing a method. State the equality case.
3. Suggest a check of the result by a second method, from a square or a study of variation.

Exercise 9.7 – Uniqueness under a hypothesis

Extension. Assume that the exponential function has already been constructed. Let f be a function differentiable on $\mathbb{R}$ that satisfies $f\prime(x) = f(x)$ for every real number x and $f(0) = 1$. Let $g(x) = f(x)e^{-x}$.

1. Calculate $g\prime$ and explain why g is constant on $\mathbb{R}$.
2. Determine this constant, then deduce the expression of f.
3. Does this proof establish the existence or the uniqueness of the function sought? Justify.

Exercise 9.8 – Delayed review

Define the sequence (u_n) by $u_n = e^{2n+1}$ for every natural number n.

1. Prove that this sequence is geometric, then state its first term and common ratio.
2. Calculate $u_0 + u_1 + u_2$ in exact form. Compare the direct calculation with the use of a sum formula.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 187, questions a, b and c. Simplify the products and quotients, citing the property of exponentials used.

CQFD, exercise 30 p. 187, questions a, b and c. Solve the three inequalities and write the solution sets as intervals.

MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $e^0 =$
   A. 0
   B. 1
   C. e
   D. -1
2. $e^x e^y =$
   A. e^{x+y}
   B. e^{xy}
   C. $e^x + e^y$
   D. e^{x-y}
3. Derivative of e^{-2x}:
   A. e^{-2x}
   B. $2e^{-2x}$
   C. $-e^{-2x}$
   D. $-2e^{-2x}$
4. $e^x = 1$ is equivalent to
   A. $x = 1$
   B. $x = e$
   C. $x = 0$
   D. no solution
5. The sign of e^{-x} is
   A. negative if x>0
   B. always strictly positive
   C. zero at 0
   D. variable

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $e^{2x} < e^3$ is equivalent to
   A. $x < 3$
   B. $x > \frac{3}{2}$
   C. $x < \frac{3}{2}$
   D. $x < \sqrt{3}$
2. $u_n = e^{3n}$ has common ratio
   A. e^3
   B. 3

C. e

D. $3e$

3. Tangent to e^x at 0:

A. $y = x$

B. $y = ex$

C. $y = 1$

D. $y = x + 1$

4. Derivative of xe^x:

A. e^x

B. $(x + 1)e^x$

C. xe^x

D. xe^{x-1}

5. $e^x \geq 1 + x$ has equality

A. for all x

B. at 1 only

C. at 0 only

D. never

True or false? Justify.

1. $e^{2x} = 2e^x$ for all x.

2. The function $t \mapsto e^{-3t}$ is decreasing.

3. A continuous rate k=0.1 corresponds exactly to a 10 % increase in one unit of time.

Self-assessment

I keep the rules for powers; I know the positivity; I differentiate with the factor; I separate model, exact calculation and simulation.
For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.

Solutions and teaching notes for chapter 9

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $2^7 = 128$. *Action:* Rebuild the product of seven factors.
2. $\frac{1}{9}$. *Action:* Go back to the reciprocal power.
3. $x = 2$. *Action:* Isolate the unknown while preserving equivalence.
4. $2x + 2$. *Action:* Apply the sum of the derivatives.
5. 0.8. *Action:* Subtract the rate from 1.
6. 5×2^n. *Action:* Count the n multiplications.

Investigation activity

The condition $f\prime = f$ motivates $\Delta y \approx hy$. With h=$\frac{1}{4}$, the values are 1; 1.25; 1.5625; 1.953125. This construction approximates the function and does not replace its characterisation.

Immediate applications

1. $y = e + e(x - 1) = ex$.
2. e^{2x+1}; $x + 2 > 2x$, so $x < 2$.
3. $g\prime(t) = -6e^{-2t} < 0$ on $\mathbb{R}$: strictly decreasing.
4. $u_n = 5(e^{0.3})^n$; the ratio over two units is $e^{0.6}$.
5. Apply the inequality proved with the argument $2x$; equality only at x=0.

Exercises: method and results

Exercise 9.1 – Transforming. e^2; e^{5x}; $1 + e^{-x}$. For the last expression, distribute then simplify the products.

Exercise 9.2 – Equations and inequalities. By strict increase: $2x - 1 = x + 3$, so x=4. Then $-3x \geq x - 2$, that is $x \leq \frac{1}{2}$.

Exercise 9.3 – Product and maximum. $f\prime = (1 - x)e^{-x}$; increasing up to 1 then decreasing. Maximum $f(1) = \frac{1}{e}$. Values at the endpoints 0 and $4e^{-4}$ are smaller.

Exercise 9.4 – A modelled population. 200; factor $e^{0.1}$; rate $e^{0.1} - 1 \approx 10.52\,\%$. Resources and the carrying capacity of the environment are not taken into account; this is a model over a limited time. Both curves requested pass through (0;1); the first increases, the second decreases. They are symmetric about the vertical axis; see the figure in the section From discrete to continuous.

Exercise 9.5 – Searching for a threshold. Test n=0,1,... in turn with an iteration limit. $Q(4) \approx 10.8268 \geq 10$, $Q(5) \approx 6.5668 < 10$; decrease proved, so the first integer is 5.

Exercise 9.6 – A symmetry. $f(-x) = f(x)$: even. $f' = e^x - e^{-x}$ has the sign of x, so minimum 2 at 0. Another proof: $f(x) - 2 = (e^{\frac{x}{2}} - e^{-\frac{x}{2}})^2$.

Exercise 9.7 – Uniqueness under a hypothesis. $g' = f'e^{-x} - fe^{-x} = 0$. On $\mathbb{R}$, g is constant, and g(0)=1. Thus $f(x) = e^x$. This proof of uniqueness uses an exponential already constructed; it does not prove existence.

Exercise 9.8 – Delayed review. $\frac{u_{n+1}}{u_n} = e^2$; first term e. Sum $e(1 + e^2 + e^4) = \frac{e(1 - e^6)}{1 - e^2}$. The direct sum is just as efficient here.

CQFD, exercise 18 p. 187, questions a, b and c. a: e^{3x}. b: e^2, because e^{x+2} is divided by e^x, which is never zero. c: e^{x+1}. Do not confuse the number e with the variable x.

CQFD, exercise 30 p. 187, questions a, b and c. a: $x \leq 0$. b: $2x > 1$, so $x > \frac{1}{2}$. c: $-3x \geq 2x + 1$, so $x \leq -\frac{1}{5}$. The strict increase of the exponential preserves the direction; dividing by -5 reverses it in c.

MCQ: answers and reasons

MCQ 1.

1. B. Initial condition.
2. A. Addition property.
3. D. Factor -2.
4. C. Strict increase and $e^0 = 1$.
5. B. Positivity of the exponential.

MCQ 2.

1. C. Compare the exponents.
2. A. Ratio of consecutive terms.
3. D. Value 1 and slope 1.
4. B. Product rule.
5. C. Minimum of the difference.

True or false

1. False: at 0, 1 is different from 2.
2. True: its derivative $-3e^{-3t}$ is negative.
3. False: the discrete rate is $e^{0.1} - 1$, about 10.52 %.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.